Modeled Re-entry Telemetry — Orion Lunar-Return Skip Entry

Entry Interface (EI) → Splashdown. Physics-based reference model (see notes at bottom). MET = Mission Elapsed Time measured from Entry Interface.

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
0:00	24,592	400,000	75.8		Entry Interface (EI) - 400,000 ft, atmosphere begins
0:10	24,791	383,333	72.6	0.9	Entry Interface (EI) - 400,000 ft, atmosphere begins
0:20	24,991	366,667	69.4	0.9	Entry Interface (EI) - 400,000 ft,

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					atmosphere begins
0:30	25,190	350,000	66.3	0.9	Initial descent - still accelerating slightly under gravity
0:40	24,948	331,111	62.7	1.1	Initial descent - still accelerating slightly under gravity
0:50	24,706	312,222	59.1	1.1	Initial descent - still accelerating slightly

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					under gravity
1:00	24,463	293,333	55.6	1.1	Initial descent - still accelerating slightly under gravity
1:10	24,221	274,444	52.0	1.1	Initial descent - still accelerating slightly under gravity
1:15	24,100	265,000	50.2	1.1	Atmospheric drag building
1:20	23,443	258,571	49.0	6.0	Atmospheric drag

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					building
1:30	22,129	245,714	46.5	6.0	Atmospheric drag building
1:40	20,814	232,857	44.1	6.0	Atmospheric drag building
1:50	19,500	220,000	41.7	6.0	First peak heating / peak-g (first entry)
2:00	18,250	216,250	41.0	5.7	First peak heating / peak-g (first entry)
2:10	17,000	212,500	40.2	5.7	First peak heating / peak-g (first entry)
2:20	15,750	208,750	39.5	5.7	First

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					peak heating / peak-g (first entry)
2:30	14,500	205,000	38.8	5.7	Comms blackout - plasma sheath (first entry)
2:40	13,930	206,000	39.0	2.6	Comms blackout - plasma sheath (first entry)
2:50	13,360	207,000	39.2	2.6	Comms blackout - plasma sheath (first entr

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					y)
3:00	12,790	208,000	39.4	2.6	Comms blackout - plasma sheath (first entry)
3:10	12,220	209,000	39.6	2.6	Comms blackout - plasma sheath (first entry)
3:20	11,650	210,000	39.8	2.6	End of first entry - velocity bled off
3:30	11,106	215,556	40.8	2.5	End of first entry - velocity bled

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					off
3:40	10,561	221,111	41.9	2.5	End of first entry - velocity bled off
3:50	10,017	226,667	42.9	2.5	End of first entry - velocity bled off
4:00	9,472	232,222	44.0	2.5	End of first entry - velocity bled off
4:05	9,200	235,000	44.5	2.5	SKIP / loft - vehicle climbs, drag drops
4:10	9,118	236,818	44.9	0.7	SKIP / loft - vehi

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					cle climbs, drag drops
4:20	8,955	240,455	45.5	0.7	SKIP / loft - vehicle climbs, drag drops
4:30	8,791	244,091	46.2	0.7	SKIP / loft - vehicle climbs, drag drops
4:40	8,627	247,727	46.9	0.7	SKIP / loft - vehicle climbs, drag drops
4:50	8,464	251,364	47.6	0.7	SKIP / loft - vehicle clim

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					bs, drag drops
5:00	8,300	255,000	48.3	0.7	Skip apogee (loft peak) - thin air, low drag
5:10	8,267	252,500	47.8	0.2	Skip apogee (loft peak) - thin air, low drag
5:20	8,233	250,000	47.3	0.2	Skip apogee (loft peak) - thin air, low drag
5:30	8,200	247,500	46.9	0.2	Skip apogee (loft peak) - thin air,

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					low drag
5:40	8,167	245,000	46.4	0.2	Skip apogee (loft peak) - thin air, low drag
5:50	8,133	242,500	45.9	0.2	Skip apogee (loft peak) - thin air, low drag
6:00	8,100	240,000	45.5	0.2	Descending again toward second entry
6:10	8,029	235,000	44.5	0.3	Descending again toward second

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					entry
6:20	7,957	230,000	43.6	0.3	Descending again toward second entry
6:30	7,886	225,000	42.6	0.3	Descending again toward second entry
6:40	7,814	220,000	41.7	0.3	Descending again toward second entry
6:50	7,743	215,000	40.7	0.3	Descending again toward

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					second entry
7:00	7,671	210,000	39.8	0.3	Descending again toward second entry
7:10	7,600	205,000	38.8	0.3	Second entry - atmosphere re-engages
7:20	7,120	197,000	37.3	2.2	Second entry - atmosphere re-engages
7:30	6,640	189,000	35.8	2.2	Second entry - atmosphere

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					re-engages
7:40	6,160	181,000	34.3	2.2	Second entry - atmosphere re-engages
7:50	5,680	173,000	32.8	2.2	Second entry - atmosphere re-engages
8:00	5,200	165,000	31.2	2.2	Second peak heating (second entry)
8:10	4,767	157,500	29.8	2.0	Second peak heating (second entry)
8:20	4,333	150,000	28.4	2.0	Seco

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					nd peak heating (second entry)
8:30	3,900	142,500	27.0	2.0	Second peak heating (second entry)
8:40	3,467	135,000	25.6	2.0	Second peak heating (second entry)
8:50	3,033	127,500	24.1	2.0	Second peak heating (second entry)
9:00	2,600	120,000	22.7	2.0	Rapid deceleration - lower

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					atmosphere
9:10	2,358	115,000	21.8	1.1	Rapid deceleration - lower atmosphere
9:20	2,117	110,000	20.8	1.1	Rapid deceleration - lower atmosphere
9:30	1,875	105,000	19.9	1.1	Rapid deceleration - lower atmosphere
9:40	1,633	100,000	18.9	1.1	Rapid deceleration - lower atmosphe

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					re
9:50	1,392	95,000	18.0	1.1	Rapid deceleration - lower atmosphere
10:00	1,150	90,000	17.0	1.1	Subsonic transition approaching
10:10	1,052	83,333	15.8	0.4	Subsonic transition approaching
10:20	953	76,667	14.5	0.4	Subsonic transition approaching
10:30	855	70,000	13.3	0.4	Subsonic transition approaching
10:40	757	63,333	12.0	0.4	Subsonic trans

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					ition approaching
10:50	658	56,667	10.7	0.4	Subsonic transition approaching
11:00	560	50,000	9.5	0.4	Subsonic - stabilizing for chute sequence
11:10	505	43,750	8.3	0.3	Subsonic - stabilizing for chute sequence
11:20	450	37,500	7.1	0.3	Subsonic - stabilizing for chute sequence
11:30	395	31,250	5.9	0.3	Subs

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					onic - stabilizing for chute sequence
11:40	340	25,000	4.7	0.3	DROGUE parachutes deploy (~25,000 ft)
11:50	305	22,833	4.3	0.2	DROGUE parachutes deploy (~25,000 ft)
12:00	270	20,667	3.9	0.2	DROGUE parachutes deploy (~25,000 ft)
12:10	235	18,500	3.5	0.2	DROGUE para

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					chutes deploy (~25,000 ft)
12:20	200	16,333	3.1	0.2	DROGUE parachutes deploy (~25,000 ft)
12:30	165	14,167	2.7	0.2	DROGUE parachutes deploy (~25,000 ft)
12:40	130	12,000	2.3	0.2	Drogues stabilizing descent
12:50	118	11,167	2.1	0.1	Drogues stabilizing descent
13:00	107	10,333	2.0	0.1	Drog

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					ues stabilizing descent
13:10	95	9,500	1.8	0.1	MAIN parachutes deploy (~9,500 ft)
13:20	83	8,917	1.7	0.1	MAIN parachutes deploy (~9,500 ft)
13:30	71	8,333	1.6	0.1	MAIN parachutes deploy (~9,500 ft)
13:40	58	7,750	1.5	0.1	MAIN parachutes depl

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					oy (~9,500 ft)
13:50	46	7,167	1.4	0.1	MAIN parachutes deploy (~9,500 ft)
14:00	34	6,583	1.2	0.1	MAIN parachutes deploy (~9,500 ft)
14:10	22	6,000	1.1	0.1	Under mains - terminal descent
14:20	22	5,733	1.1	0.0	Under mains - terminal descent
14:30	22	5,467	1.0	0.0	Under

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					mains - terminal descent
14:40	22	5,200	1.0	0.0	Under mains - terminal descent
14:50	21	4,933	0.9	0.0	Under mains - terminal descent
15:00	21	4,667	0.9	0.0	Under mains - terminal descent
15:10	21	4,400	0.8	0.0	Under mains - terminal descent
15:20	21	4,133	0.8	0.0	Under mains -

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					terminal descent
15:30	21	3,867	0.7	0.0	Under mains - terminal descent
15:40	21	3,600	0.7	0.0	Under mains - terminal descent
15:50	21	3,333	0.6	0.0	Under mains - terminal descent
16:00	21	3,067	0.6	0.0	Under mains - terminal descent
16:10	20	2,800	0.5	0.0	Under mains - terminal

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					descent
16:20	20	2,533	0.5	0.0	Under mains - terminal descent
16:30	20	2,267	0.4	0.0	Under mains - terminal descent
16:40	20	2,000	0.4	0.0	Under mains - steady descent
16:50	20	1,857	0.4	0.0	Under mains - steady descent
17:00	20	1,714	0.3	0.0	Under mains - steady descent

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
17:10	19	1,571	0.3	0.0	Under mains - steady descent
17:20	19	1,429	0.3	0.0	Under mains - steady descent
17:30	19	1,286	0.2	0.0	Under mains - steady descent
17:40	19	1,143	0.2	0.0	Under mains - steady descent
17:50	18	1,000	0.2	0.0	Under mains - steady descent
18:00	18	857	0.2	0.0	Under

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					mains - steady descent
18:10	18	714	0.1	0.0	Under mains - steady descent
18:20	18	571	0.1	0.0	Under mains - steady descent
18:30	18	429	0.1	0.0	Under mains - steady descent
18:40	17	286	0.1	0.0	Under mains - steady descent
18:50	17	143	0.0	0.0	Under mains -

MET (from EI)	Velocity (mph)	Altitude (ft)	Altitude (mi)	Decel (g)	Phase
					steady descent
19:00	17	0	0.0	0.0	SPLASHDOWN - Pacific Ocean